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Ingeniería en Informática y Sistemas  
Redes y telecomunicaciones - Sección 1  
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# **REDES Y TELECOMUNICACIONES:**

## **LABORATORIO 12**

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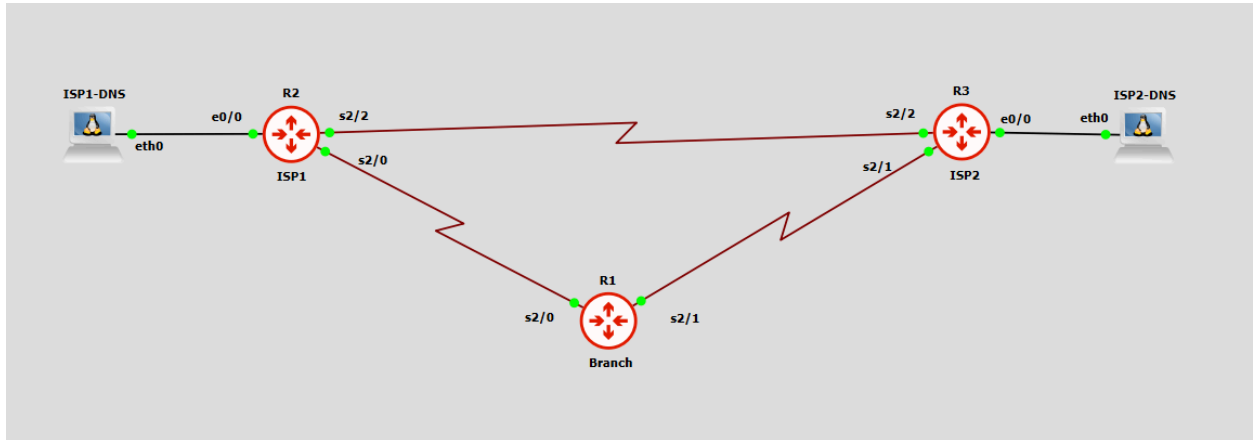
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## **ENLACE A VIDEO EXPLICATIVO**

<https://youtu.be/cAl0mfM5cmE>

# CAPTURAS DE PANTALLA PARA LA REALIZACIÓN DEL LABORATORIO

## Topología



*Nota: Se utilizaron imágenes Docker de Linux para simular los servidores DNS y el Web Server*

## Preparación de los routers y las direcciones de las interfaces

### Router 1 (Branch):

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - IISIP
       + - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

S*   0.0.0.0/0 [2/0] via 209.165.201.1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.1.0/24 is directly connected, Loopback0
L     192.168.1.1/32 is directly connected, Loopback0
C     209.165.201.0/24 is variably subnetted, 3 subnets, 2 masks
C     209.165.201.0/30 is directly connected, Serial2/0
L     209.165.201.2/32 is directly connected, Serial2/0
S     209.165.201.30/32 [1/0] via 209.165.201.1
C     209.165.202.0/24 is variably subnetted, 3 subnets, 2 masks
C     209.165.202.128/30 is directly connected, Serial2/1
L     209.165.202.130/32 is directly connected, Serial2/1
S     209.165.202.158/32 [1/0] via 209.165.202.129
R1#
```

```
R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
    Latest RTT: 9 milliseconds
Latest operation start time: 20:41:58 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 19
Number of failures: 97
Operation time to live: Forever

IPSLA operation id: 22
    Latest RTT: 9 milliseconds
Latest operation start time: 20:41:58 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 19
Number of failures: 97
Operation time to live: Forever

R1#traceroute 209.165.200.254 source 192.168.1.1
Type escape sequence to abort.
Tracing the route to 209.165.200.254
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 9 msec 8 msec 8 msec
R1#
```

```
R1#show interfaces description
Interface      Status      Protocol Description
Et0/0          admin down  down
Et0/1          admin down  down
Et0/2          admin down  down
Et0/3          admin down  down
Et1/0          admin down  down
Et1/1          admin down  down
Et1/2          admin down  down
Et1/3          admin down  down
Se2/0          up          up        R1 to ISP1
Se2/1          up          up        R1 to ISP2
Se2/2          admin down  down
Se2/3          admin down  down
Se3/0          admin down  down
Se3/1          admin down  down
Se3/2          admin down  down
Se3/3          admin down  down
Lo0            up          up        R1 LAN
```

Router 2 (ISP 1):

```
ISP1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
S      192.168.1.0/24 [1/0] via 209.165.201.2
      209.165.200.0/24 is variably subnetted, 3 subnets, 2 masks
C      209.165.200.224/30 is directly connected, Serial2/2
L      209.165.200.225/32 is directly connected, Serial2/2
C      209.165.200.254/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 4 subnets, 3 masks
C      209.165.201.0/30 is directly connected, Serial2/0
L      209.165.201.1/32 is directly connected, Serial2/0
C      209.165.201.24/29 is directly connected, Ethernet0/0
L      209.165.201.25/32 is directly connected, Ethernet0/0
ISP1#
```

```
ISP1#show interfaces description
```

| Interface | Status     | Protocol | Description                   |
|-----------|------------|----------|-------------------------------|
| Et0/0     | up         | up       | ISP1 to Docker DNS            |
| Et0/1     | admin down | down     |                               |
| Et0/2     | admin down | down     |                               |
| Et0/3     | admin down | down     |                               |
| Et1/0     | admin down | down     |                               |
| Et1/1     | admin down | down     |                               |
| Et1/2     | admin down | down     |                               |
| Et1/3     | admin down | down     |                               |
| Se2/0     | up         | up       | ISP1 to R1                    |
| Se2/1     | admin down | down     |                               |
| Se2/2     | up         | up       | ISP1 to ISP2                  |
| Se2/3     | admin down | down     |                               |
| Se3/0     | admin down | down     |                               |
| Se3/1     | admin down | down     |                               |
| Se3/2     | admin down | down     |                               |
| Se3/3     | admin down | down     |                               |
| Lo0       | up         | up       | Simulated Internet Web Server |

### Router 3 (ISP 2):

```
ISP2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
S    192.168.1.0/24 [1/0] via 209.165.202.130
    209.165.200.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.200.224/30 is directly connected, Serial2/2
L    209.165.200.226/32 is directly connected, Serial2/2
C    209.165.200.254/32 is directly connected, Loopback0
    209.165.202.0/24 is variably subnetted, 4 subnets, 3 masks
C    209.165.202.128/30 is directly connected, Serial2/1
L    209.165.202.129/32 is directly connected, Serial2/1
C    209.165.202.152/29 is directly connected, Ethernet0/0
L    209.165.202.153/32 is directly connected, Ethernet0/0
```

```
ISP2#show interfaces description
```

| Interface | Status     | Protocol | Description                   |
|-----------|------------|----------|-------------------------------|
| Et0/0     | up         | up       | ISP2 to Docker DNS            |
| Et0/1     | admin down | down     |                               |
| Et0/2     | admin down | down     |                               |
| Et0/3     | admin down | down     |                               |
| Et1/0     | admin down | down     |                               |
| Et1/1     | admin down | down     |                               |
| Et1/2     | admin down | down     |                               |
| Et1/3     | admin down | down     |                               |
| Se2/0     | admin down | down     |                               |
| Se2/1     | up         | up       | ISP2 to R1                    |
| Se2/2     | up         | up       | ISP2 to ISP1                  |
| Se2/3     | admin down | down     |                               |
| Se3/0     | admin down | down     |                               |
| Se3/1     | admin down | down     |                               |
| Se3/2     | admin down | down     |                               |
| Se3/3     | admin down | down     |                               |
| Lo0       | up         | up       | Simulated Internet Web Server |

## Dockers Linux

| ISP1-DNS interfaces                                                                                                           | ISP2-DNS interfaces                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <pre>auto eth0 iface eth0 inet static     address 209.165.201.30     netmask 255.255.255.248     gateway 209.165.201.25</pre> | <pre>auto eth0 iface eth0 inet static     address 209.165.202.158     netmask 255.255.255.248     gateway 209.165.202.153</pre> |

## Verificación de conexión

### Pings a servidores DNS

```
R1#tclsh
R1(tcl)#foreach address {
+>(tcl)#209.165.200.254
+>(tcl)#209.165.201.30
+>(tcl)#209.165.202.158
+>(tcl)#} {
+>(tcl)#ping $address source 192.168.1.1
+>(tcl)#}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.254, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/8/9 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.30, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/7/9 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.158, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/8/9 ms
R1(tcl)#
```

Trace de pings a ISP1, ISP2 y servidor web

```
R1(tcl)#foreach address {
+>(tcl)#209.165.200.254
+>(tcl)#209.165.201.30
+>(tcl)#209.165.202.158
+>(tcl)#} {
+>(tcl)#trace $address source 192.168.1.1
+>(tcl)#}
Type escape sequence to abort.
Tracing the route to 209.165.200.254
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 9 msec 8 msec 9 msec
Type escape sequence to abort.
Tracing the route to 209.165.201.30
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.201.1 9 msec 5 msec 7 msec
  2 209.165.201.30 9 msec 9 msec 8 msec
Type escape sequence to abort.
Tracing the route to 209.165.202.158
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.202.129 8 msec 8 msec 9 msec
  2 209.165.202.158 8 msec 9 msec 8 msec
R1(tcl)#
```

- ¿Por cuál ISP está pasando el tráfico?
  - El tráfico principal está pasando por ISP1, porque pingean a 209.165.201.1, la dirección que se le asignó al ISP1

## Configuración de prueba para SLA

### SLA 11

```
R1#show ip sla configuration 11
IP SLAs Infrastructure Engine-III
Entry number: 11
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 209.165.201.30/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
  Operation frequency (seconds): 10 (not considered if randomly scheduled)
  Next Scheduled Start Time: Start Time already passed
  Group Scheduled : FALSE
  Randomly Scheduled : FALSE
  Life (seconds): Forever
  Entry Ageout (seconds): never
  Recurring (Starting Everyday): FALSE
  Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
  Number of statistic hours kept: 2
  Number of statistic distribution buckets kept: 1
  Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
  Number of history Lives kept: 0
  Number of history Buckets kept: 15
  History Filter Type: None
```

```
R1#sho ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
  Latest RTT: 9 milliseconds
Latest operation start time: 21:17:03 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 3
Number of failures: 0
Operation time to live: Forever

IPSLA operation id: 22
  Latest RTT: 9 milliseconds
Latest operation start time: 21:16:58 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 229
Number of failures: 97
Operation time to live: Forever
```

## SLA 22

```
R1#show ip sla conf
R1#show ip sla configuration 22
IP SLAs Infrastructure Engine-III
Entry number: 22
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 209.165.202.158/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
  Operation frequency (seconds): 10 (not considered if randomly scheduled)
  Next Scheduled Start Time: Start Time already passed
  Group Scheduled : FALSE
  Randomly Scheduled : FALSE
  Life (seconds): Forever
  Entry Ageout (seconds): never
  Recurring (Starting Everyday): FALSE
  Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
  Number of statistic hours kept: 2
  Number of statistic distribution buckets kept: 1
  Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
  Number of history Lives kept: 0
  Number of history Buckets kept: 15
  History Filter Type: None
```

```
R1#show ip sla statistics 22
IPSLAs Latest Operation Statistics

IPSLA operation id: 22
  Latest RTT: 9 milliseconds
Latest operation start time: 21:21:58 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 259
Number of failures: 97
Operation time to live: Forever
```

## Configuración de opciones de monitoreo

Configuración de ruta con distancia de 5

```
S* 0.0.0.0/0 [2/0] via 209.165.201.1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, Loopback0
L    192.168.1.1/32 is directly connected, Loopback0
    209.165.201.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.201.0/30 is directly connected, Serial2/0
L    209.165.201.2/32 is directly connected, Serial2/0
S    209.165.201.30/32 [1/0] via 209.165.201.1
    209.165.202.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.202.128/30 is directly connected, Serial2/1
L    209.165.202.130/32 is directly connected, Serial2/1
S    209.165.202.158/32 [1/0] via 209.165.202.129
```

```
R1#show run | include ip route
ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1
ip route 0.0.0.0 0.0.0.0 209.165.202.129 3 track 2
ip route 0.0.0.0 0.0.0.0 209.165.201.1 5
ip route 209.165.201.30 255.255.255.255 209.165.201.1
ip route 209.165.202.158 255.255.255.255 209.165.202.129
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

```
S* 0.0.0.0/0 [2/0] via 209.165.201.1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, Loopback0
L    192.168.1.1/32 is directly connected, Loopback0
    209.165.201.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.201.0/30 is directly connected, Serial2/0
L    209.165.201.2/32 is directly connected, Serial2/0
S    209.165.201.30/32 [1/0] via 209.165.201.1
    209.165.202.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.202.128/30 is directly connected, Serial2/1
L    209.165.202.130/32 is directly connected, Serial2/1
--More--
*Apr 19 21:44:20.781: %SYS-5-CONFIG_I: Configured from console by console
S    209.165.202.158/32 [1/0] via 209.165.202.129
```

## Prueba de deshabilitación de ISP1-DNS

### Deshabilitación de ISP1-DNS

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#track 2 ip sla 22 reachability
R1(config-track)#delay down 10 up 1
R1(config-track)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.202.129 3 track 2
R1(config)#
R1(config)#
R1(config)#track 2 ip sla 22 reachability
R1(config-track)#delay down 10 up 1
R1(config-track)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.202.129 3 track 2
R1(config)#exit
R1#
*Apr 19 21:32:45.412: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
*Apr 19 21:33:31.067: %TRACKING-5-STATE: 1 ip sla 11 reachability Up->Down
*Apr 19 21:33:31.067: RT: del 0.0.0.0 via 209.165.201.1, static metric [2/0]
*Apr 19 21:33:31.067: RT: delete network route to 0.0.0.0/0
*Apr 19 21:33:31.067: RT: default path has been cleared
*Apr 19 21:33:31.067: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.202.129
*Apr 19 21:33:31.067: RT: add 0.0.0.0/0 via 209.165.202.129, static metric [3/0]
*Apr 19 21:33:31.067: RT: default path is now 0.0.0.0 via 209.165.202.129
*Apr 19 21:33:31.067: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.201.1
*Apr 19 21:33:31.067: RT: rib update return code: 17
*Apr 19 21:33:31.067: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.202.129
R1#
*Apr 19 21:33:31.067: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.201.1
*Apr 19 21:33:31.067: RT: rib update return code: 17
R1#
```

Gateway of last resort is 209.165.202.129 to network 0.0.0.0

```
S*    0.0.0.0/0 [3/0] via 209.165.202.129
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback0
L      192.168.1.1/32 is directly connected, Loopback0
      209.165.201.0/24 is variably subnetted, 3 subnets, 2 masks
C      209.165.201.0/30 is directly connected, Serial2/0
L      209.165.201.2/32 is directly connected, Serial2/0
S      209.165.201.30/32 [1/0] via 209.165.201.1
      209.165.202.0/24 is variably subnetted, 3 subnets, 2 masks
C      209.165.202.128/30 is directly connected, Serial2/1
L      209.165.202.130/32 is directly connected, Serial2/1
S      209.165.202.158/32 [1/0] via 209.165.202.129
```

```
R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
    Latest RTT: NoConnection/Busy/Timeout
Latest operation start time: 21:47:03 UTC Sun Apr 19 2026
Latest operation return code: Timeout
Number of successes: 148
Number of failures: 35
Operation time to live: Forever

IPSLA operation id: 22
    Latest RTT: 9 milliseconds
Latest operation start time: 21:47:08 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 147
Number of failures: 0
Operation time to live: Forever

R1#trace 209.165.200.254 source 192.168.1.1
Type escape sequence to abort.
Tracing the route to 209.165.200.254
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.202.129 9 msec 8 msec 9 msec
R1#
```

## Rehabilitación de ISP1-DNS

```
R1#
*Apr 19 21:50:47.566: %TRACKING-5-STATE: 1 ip sla 11 reachability Down->Up
R1#
*Apr 19 21:50:47.566: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.201.1
*Apr 19 21:50:47.566: RT: closer admin distance for 0.0.0.0, flushing 1 routes
*Apr 19 21:50:47.566: RT: add 0.0.0.0/0 via 209.165.201.1, static metric [2/0]
*Apr 19 21:50:47.566: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.202.129
*Apr 19 21:50:47.566: RT: rib update return code: 17
*Apr 19 21:50:47.566: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.202.129
*Apr 19 21:50:47.567: RT: rib update return code: 17
*Apr 19 21:50:47.567: RT: updating static 0.0.0.0/0 (0x0):
    via 209.165.201.1
*Apr 19 21:50:47.567: RT: rib update return code: 17
```

```
R1#show ip sla statistics
IPSLAs Latest Operation Statistics
```

```
IPSLA operation id: 11
    Latest RTT: 9 milliseconds
Latest operation start time: 21:51:33 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 154
Number of failures: 56
Operation time to live: Forever
```

```
IPSLA operation id: 22
    Latest RTT: 9 milliseconds
Latest operation start time: 21:51:28 UTC Sun Apr 19 2026
Latest operation return code: OK
Number of successes: 173
Number of failures: 0
Operation time to live: Forever
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override

Gateway of last resort is 209.165.201.1 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [2/0] via 209.165.201.1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, Loopback0
L    192.168.1.1/32 is directly connected, Loopback0
    209.165.201.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.201.0/30 is directly connected, Serial2/0
L    209.165.201.2/32 is directly connected, Serial2/0
S    209.165.201.30/32 [1/0] via 209.165.201.1
    209.165.202.0/24 is variably subnetted, 3 subnets, 2 masks
C    209.165.202.128/30 is directly connected, Serial2/1
L    209.165.202.130/32 is directly connected, Serial2/1
S    209.165.202.158/32 [1/0] via 209.165.202.129
```